

O-, E- and S-band Silicon Photonics Wavelength Cross-Connect Switch with Waveguide Bragg Gratings

Kazuhiro Ikeda, Ryotaro Konoike, Keiji Suzuki

Photonics and Electronics Integration Research Center, National Institute of Advanced Industrial Science and Technology, kaz.ikeda@aist.go.jp

Abstract We demonstrate an extended multiband operation over 300 nm using an improved wavelength cross-connect switch circuit in 2x2 and 32-ch scale, fully exploiting the potential available wavelength range of wavelength-selective couplers with waveguide sidewall gratings. ©2026 The Author(s)

Introduction

Silicon photonics switches are a promising technology offering highly integrated and high-speed optical switching [1][2] for high-capacity and low-power consumption datacentre networks and AI/ML clusters. To increase bandwidth density to the order of Tbps/mm or higher, many-channel wavelength division multiplexing (WDM) is essential [3], and consequently, wavelength cross-connect (WXC) switches that can flexibly switch paths for a large number of wavelength channels across a wide wavelength range will become an important technology [4]-[6]. Conventional integrated WDM filters, such as AWGs, Mach-Zehnder interferometers (MZIs), and micro ring resonators, utilize higher-order diffraction, interference, and resonance, respectively, and their free spectral ranges (FSRs) limit the available bandwidth. We proposed a novel approach [7][8] using a wavelength-selective coupler (WSC) with waveguide Bragg gratings, which has practically unlimited FSR, and a large-scale MZI-based silicon photonics switch that we have been developing. We demonstrated a 16×16, 16-wavelength channel WXC switch [9][10] using AIST's 300-mm silicon photonics prototyping line, in which the 11×26 mm WXC chip integrated 256 WSCs and 1,024 MZIs (2,048 microheaters) and the 16 wavelength channels distributed over the C-band and L-band (~90 nm). However, the datacentre networks and AI/ML clusters use the O-band and the potential available wavelength range of our WXC has not yet fully demonstrated.

In this paper, we demonstrate the silicon photonics WXC for a much wider wavelength range over the O-, E- and S-band (~300 nm), using an improved WXC circuit topology to fully exploit the potential available wavelength range of the WSC.

Silicon Wavelength-Selective Couplers

The WSC consists of two coupled waveguides with periodic refractive index modulation, resulting in counter directional mode coupling between the waveguides at the Bragg wavelength [11]. Due to its first-order operation, the FSR is in practice unlimited, making it suitable for multiband WDM filters. The silicon WSC was first proposed

in Ref. [12], where gratings on the silicon waveguide sidewalls induce large refractive index modulation (i.e., coupling coefficient) and realize a broadband and flat spectral shape. This makes it suitable for low-cost many-channel WDM systems with high tolerance to wavelength variations and temperature changes.

We discussed design considerations to extend the operation wavelength range of the WXC over the full communication band in Ref. [9], one of which is the higher order modes in the WSCs. As shown in Fig.1, if the wider waveguide of a WSC has the TE1 mode, another counter directional mode coupling between that TE1 mode and the TE0 mode of the narrower waveguide (TE0-TE1) occurs in addition to the intended mode coupling between each TE0 mode (TE0-TE0). Furthermore, coupling to the free space mode (TE0-Free Space) occurs as also shown in Fig.1. The Bragg wavelengths λ for these mode couplings are given by,

$$\lambda = (n_{\text{eff}1} + n_{\alpha})\Lambda, \quad \alpha \in \{\text{eff}2, \text{clad}\} \quad (1)$$

where Λ is the grating period, $n_{\text{eff}1}$ and $n_{\text{eff}2}$ are the effective indexes of the two waveguides, and n_{clad} is the cladding refractive index multiplied by a ratio to the waveguide direction. Since $n_{\text{eff}2}$ of TE1 is smaller than that of TE0 and n_{clad} is even smaller, the Bragg wavelengths of TE0-Free Space, TE0-TE1, and TE0-TE0 are in this order. In the previously proposed WXC circuit [7]-[9] depicted in Fig.2(a), where all the wavelength channels pass all the WSCs, such additional mode

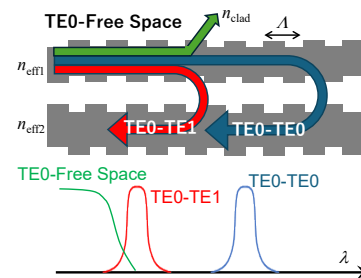


Fig. 1: Schematic illustrations of a WSC with a grating period Λ and mode couplings from the TE0 mode of the narrower waveguide with an effective index $n_{\text{eff}1}$ to the TE0 and TE1 modes of the wider waveguide with $n_{\text{eff}2}$, and to the free space mode with n_{clad} .

couplings outside the intended channel band disturb proper operation at the other wavelength channels. We carefully studied the WSC design to place the mode couplings to the higher modes and free space mode at the far shorter wavelength and to realize WXC operation for the full communication band but found it difficult. Here, we propose an improved WXC circuit exploiting the fact that the couplings to the higher order modes and free space mode always occur at the shorter wavelengths than the intended channel wavelength and the FSR is still unlimited in the longer wavelength side.

Improved WXC Circuit for Ultrawide Wavelength-Band Operation

Fig. 2(a) explains the previously proposed $N \times N$ N_λ -channel WXC circuit, where two sets of 4-channel WDM signals are introduced into the input ports labelled as 1 and 2 and guided respectively to 4 different output ports. For example, the channel described by the solid blue arrow is dropped by the first WSC designed for λ_1 and switched to the second output port from the top in the $N \times N$ switch also designed for λ_1 , then guided to the upper waveguide through another WSC. In the same way, the other 3 channels described by the solid red, green and yellow arrows are switched and guided to the other 3 waveguides. The other set of WDM signals described by the dotted arrows are also guided differently at the

same time as described in the same figure.

Fig. 2(b) explains the improved WXC circuit and the same operation as Fig. 2(a), where two modifications are implemented: (i) The waveguides for the input ports and the output ports are separated. (ii) Single-wavelength switch sections (dotted squares) are arranged from the left in order from short to long wavelength (any order is allowed in Fig. 2(a)). With these modifications, at each WSC, wavelengths shorter than the wavelength for the intended TE₀-TE₀ coupling are dropped before the WSC and do not reach the WSC (only longer wavelengths pass through). Therefore, any additional mode couplings at shorter wavelengths never occur and the unlimited FSR in the longer wavelength side of WSC enables an ultra-wideband operation of the WXC.

Design, Fabrication, and Evaluation of 2 x 2 32-Channel Silicon Photonics WXC

We designed and fabricated a 2 x 2 32-channel silicon photonics WXC with the improved circuit described in Fig. 2(b). 32 single-wavelength switch sections were concatenated from the input side in order from short to long wavelength and designed to cover the full communication band (O, E, S, C, L and U-band), each of which consisted of 2 input waveguides with 2 WSCs, a 2 x 2 thermo-optic MZI switch, and 2 output waveguides with 2 WSCs (See Fig. 3(a)).

The fabrication was performed on a 300-mm silicon-on-insulator wafer with a 220-nm silicon layer and a 3- μ m buried oxide layer using our prototyping line with 45-nm ArF immersion lithography. Figs. 3(a) and (b) show a micrograph of a single-wavelength switch section and a photo of the chip die- and wire-bonded to a chip carrier and attached by a fibre array. Note that the chip contains devices not reported in this paper and the

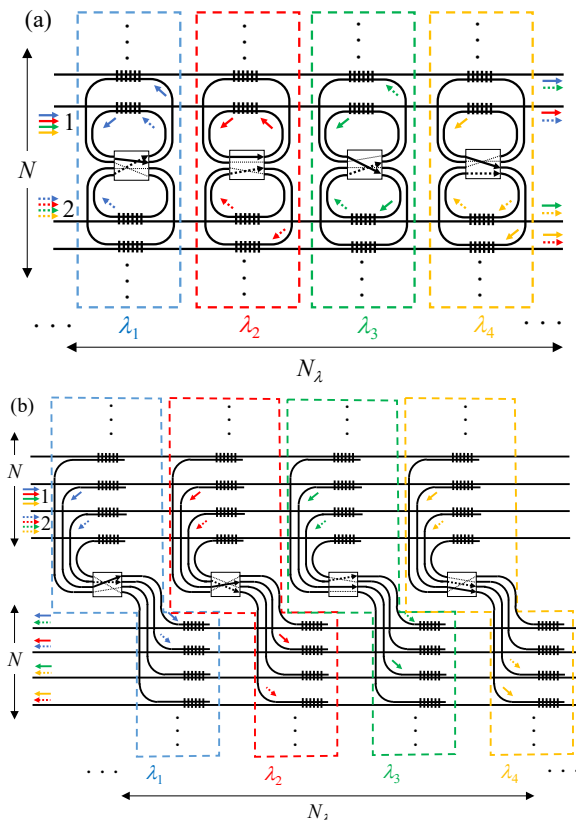


Fig. 2: Schematic illustrations of (a) the previously proposed and (b) the improved $N \times N$ N_λ -channel WXC circuits.

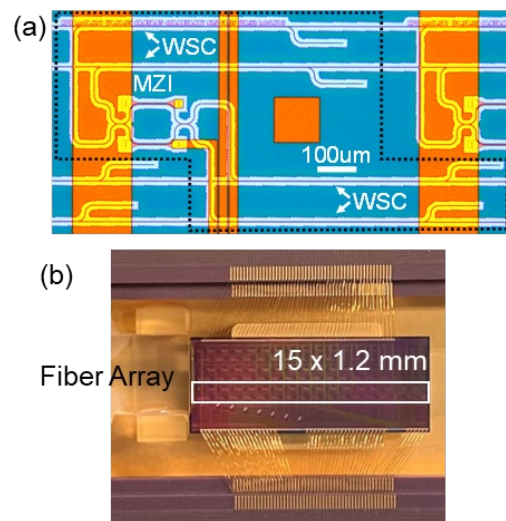


Fig. 3: (a) Micrograph of a single-wavelength 2 x 2 switch section (surrounded by the black dotted line) and (b) photo of the chip mounted on a chip carrier. The area of the 2 x 2 32-channel WXC circuit is indicated by the white square.

area of the 2×2 32-channel WXC circuit is indicated by the white square (15×1.2 mm).

An ultra broadband light source obtained by fibre-coupling multiple amplified spontaneous emission sources (Pr, Tm, and Er-doped fibre amplifiers, semiconductor optical amplifier and super luminescent diode (SLD)) for O to U-band was polarized to the transverse electric (TE) mode and introduced to the input port of the device under test. The output port was connected to an optical spectrum analyser. We first evaluated a single WSC test device to confirm the design. Fig. 4 shows measured spectra of the through and drop ports of WSC designed for TE0-TE0 counter directional mode coupling at 1510 nm. We observed 5 mode couplings at wavelengths shorter than the intended 1510 nm (more than explained in Fig.1) and they correspond to TE0-Free Space, TE0-TE1, TE0-TM0 (Self-coupling within the same waveguide), TE0-TM0, TE0-TE0 (Self) in order from the shortest wavelength according to the mode analysis. Note that the dip at around 1660 nm was caused by misalignment between the SLD light polarization and the polarizer. As explained before, these additional mode couplings at the wavelengths shorter than the intended channel wavelength do not affect the performance of WXC with the improved circuit. Fig. 5 shows measured spectra of the Cross (red) and Bar (blue) ports (extracted 21 channels over O, E and S-band) for the 2×2 32-channel silicon photonics WXC at switch states of all-cross, all-bar, Cross for O- and S-band and Bar for E-band, and interleave. Also plotted (Black) is the measured

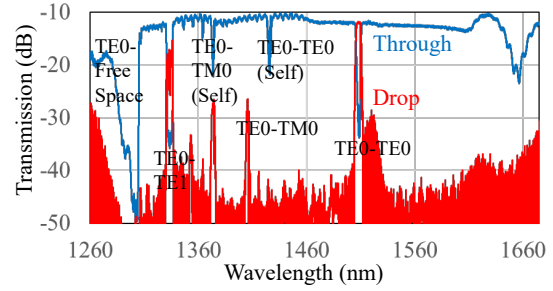


Fig. 4: Measured spectra of through and drop ports of a single WSC designed for TE0-TE0 counter directional mode coupling at 1510 nm.

spectrum for the silicon waveguide surrounding the whole WXC circuit. Since the propagation length and the number of passing components increase as the wavelength as can be seen in Fig.2(b), the loss rapidly increased toward the C, L and U-band (not shown in Fig. 5). The main loss sources are considered waveguide intersections (WIs) [9] and WSCs. The WIs used here were optimized for C-band and will be the largest loss source in this wavelength range. The lower bar port transmission than the cross port observed in Fig. 5 support this since the number of WIs were always larger in the bar port paths. For the C, L and U-band, while the loss of the WIs should be very small, the narrow waveguides (~ 300 nm) with sidewall gratings in the WSC designed for O, E and S-band are considered to cause relatively larger loss. These observations will be further analysed and used to improve the loss and to achieve operation in the full communication band in our next work.

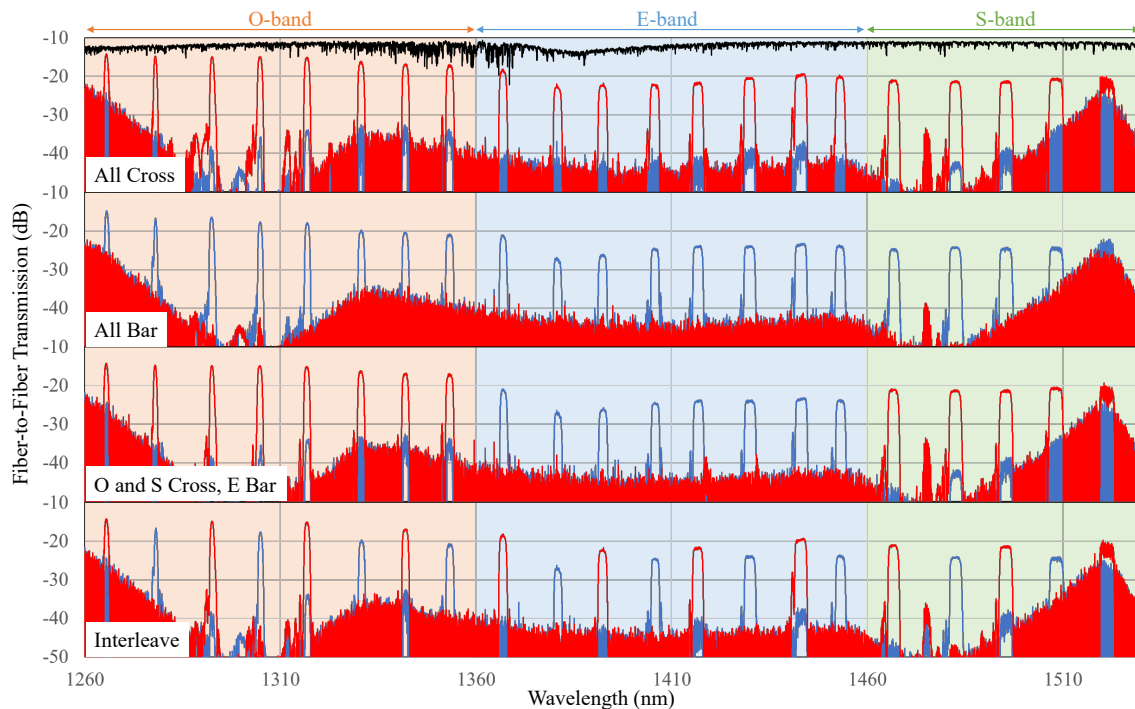


Fig. 5: Measured spectra (Red: Cross port, Blue: Bar port, Black: Straight silicon waveguide on the same chip) over O, E and S-band for 2×2 32-channel silicon photonics WXC at switch states of all-cross, all-bar, OS-cross-E-bar, and interleave.

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References

- [1] K. Ikeda, K. Suzuki, R. Konoike, S. Namiki, and H. Kawashima, "Large-scale silicon photonics switch based on 45-nm CMOS technology," *Optics Communications* 466, 125677 (2020). DOI: 10.1016/j.optcom.2020.125677
- [2] R. Matsumoto, R. Konoike, K. Suzuki, H. Matsuura, K. Ikeda, T. Inoue, and S. Namiki, "Fully-Loaded Operation of 0.29-pJ/bit Wall-plug Efficiency, 81.9-Tb/s Throughput 32×32 Silicon Photonics Switch," *Optical Fiber Communication Conference (OFC) 2021, Tu6A.2*. DOI: 10.1364/OFC.2021.Tu6A.2
- [3] <https://oci-msa.org/>
- [4] N. Harris, "Passage: A Wafer-Scale, Programmable Photonic Communication Substrate," *Hot Chips 34, Session 2, Aug 2022*. <https://hc34.hotchips.org/>
- [5] G. Van Steenberge, G. Roelkens, P. Ossieur, J. Missinne, J. Van Asch, J. Zhang, S. Qin, J. Van Campenhout, J. Van Kerkhof, M. M. Milosevic, P. Bakopoulos, D. Syrivelis, E. Mentovich, K. Morozov, S. Stracca, A. Bongiarri, T. Joerg, D. Schlick, T. Braun, M. Wöhrmann, R. Gernhardt, Q. Cheng, T. Li, and R. Penty, "Packaging of Ultra-dynamic Photonic Switches and Transceivers for Integration into 5G Radio Access Network and Datacenter Sub-systems," *IEEE 73rd Electronic Components and Technology Conference (ECTC 2023)*, p. 742. Orlando, FL, May 2023. DOI: 10.1109/ECTC51909.2023.00129
- [6] Y. Thonnart, C. Bernard, S. Bernabé, M. Assous, L. Boutafa, B. Charbonnier, R. Franiatte, C. Fuguet, O. Guille, V. Josselin, S. Malhouitre, L. Mendizabal, A. Myko, D. Saint Patrice, R. Vélard, J. Charbonnier, "A 3.19pJ/b Electro-Optical Router with 18ns Setup Frame-Level Routing and 1-to-6 Wavelength-Flexible Link Capacity for Photonic Interposers," *2026 IEEE International Solid-State Circuits Conference (ISSCC)*, 18.1, San Francisco, CA, USA, 2026, pp. 1-3. DOI: 10.1109/ISSCC49663.2026.11409067
- [7] K. Ikeda, K. Suzuki, R. Konoike, H. Kawashima, "Silicon Photonics Wavelength Selective Switch with Unlimited Free Spectral Range," *Journal of Lightwave Technology* 38, 3268-3272 (2020). DOI: 10.1109/JLT.2020.2989379
- [8] K. Ikeda, R. Konoike, K. Suzuki, H. Kawashima, " 2×2 16-ch silicon photonics wavelength-selective switch based on waveguide gratings," *Optics Express* 28, 26861-26869 (2020). DOI: 10.1364/OE.402546
- [9] K. Ikeda, R. Konoike, K. Suzuki, "Large-Scale and Multi-band Silicon Photonics Wavelength Cross-Connect Switch With FSR-Free Grating-Assisted Contra-Directional Couplers," *Journal of Lightwave Technology* 42, 4310 – 4316 (2024). DOI: 10.1109/JLT.2024.3366553
- [10] K. Ikeda, R. Matsumoto, R. Konoike and K. Suzuki, "Ultra-Wideband Wavelength Cross-Connect Switch Based on 300-mm Silicon Photonics," *Journal of Lightwave Technology* 43, 6114-6119 (2025). DOI: 10.1109/JLT.2025.3575100
- [11] P. Yeh and H. F. Taylor, "Contradirectional frequency-selective couplers for guided-wave optics," *Applied Optics* 19, 2848-2855 (1980). DOI: 10.1364/AO.19.002848
- [12] K. Ikeda, M. Nezhad, and Y. Fainman, "Wavelength selective coupler with vertical gratings on silicon chip,"